

Project Evaluation

CSE 3582 - TOPIC 2.2

Mr. Madusanka Nipunajith

Project Evaluation

A higher-level assessment of the project to see whether it is worthwhile to proceed with the project.

1. Strategic Assessment

- To see whether the project will fit in the strategic planning of the whole organization.

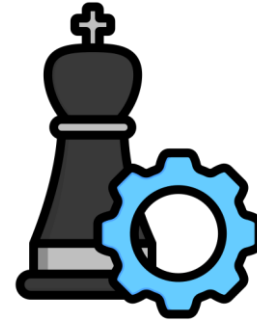
2. Technical Assessment

- To determine whether it is desirable to carry out the development and operation of the software system.

3. Economical Assessment

- To decide which of the several alternative projects has a better success rate and a high turnover.

Strategic Assessment

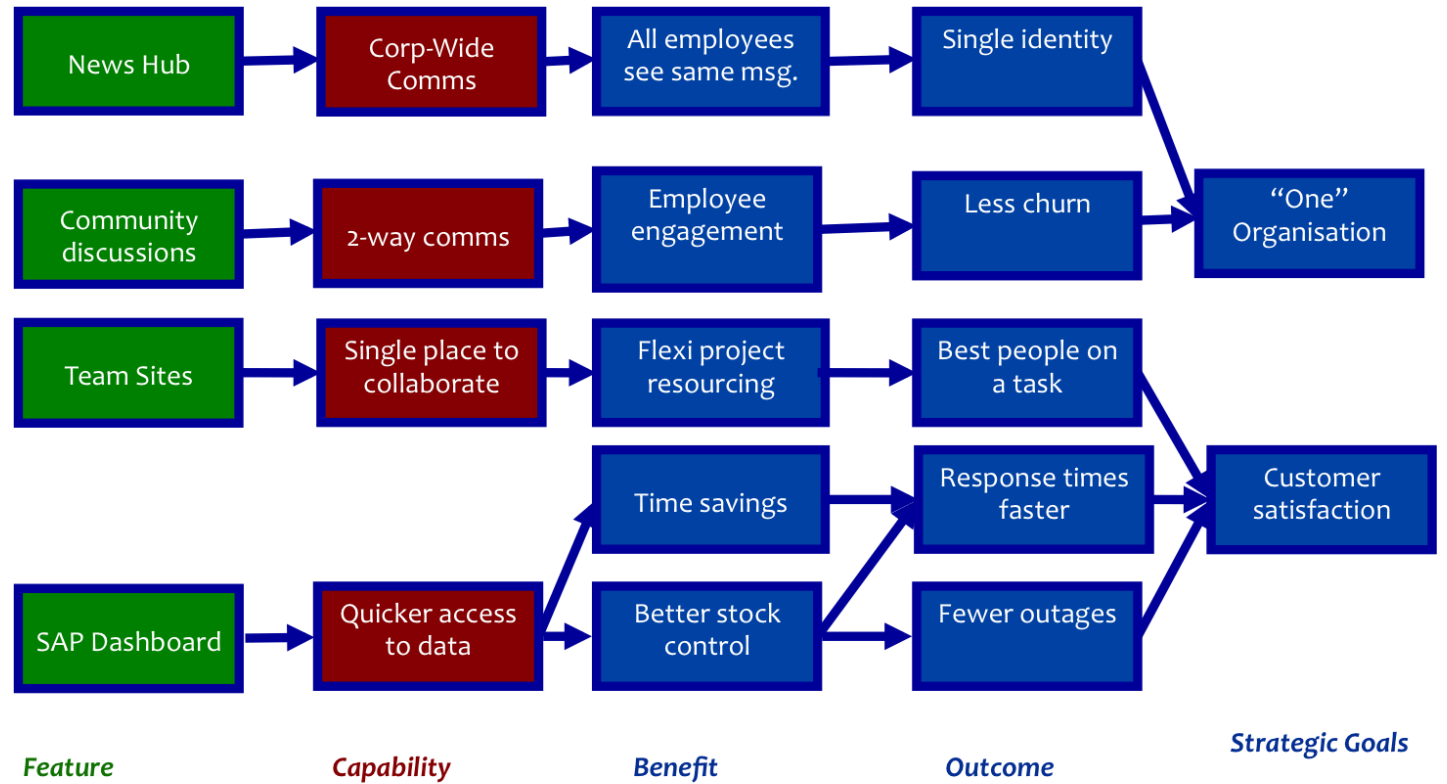


- What does it include?
 - Alignment with business strategy
 - Does the project support the org's mission, vision, and goals?.
 - Value addition
 - Will it improve competitiveness, reputation, or market share?.
 - Stakeholder alignment
 - Does it satisfy the key stakeholders (customers, investors, and partners)
 - Risk vs Rewards at an organizational level
 - Are the strategic benefits worth the potential risks

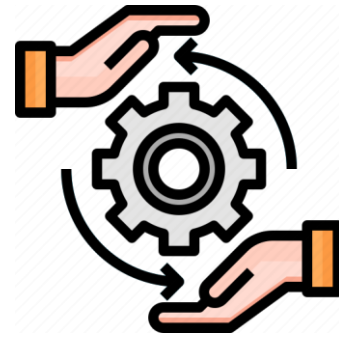
Benefit Map

- A Benefit Map is a visual tool used in project evaluation and business analysis to show how a project's activities and outcomes lead to strategic benefits for the organization.
- A Benefit Map links together:
 - Project Activities/Changes
 - What you're doing in the project (e.g., build new software/feature)
 - Enablers/Capabilities
 - What the project enables (e.g., faster data processing).
 - Business Benefits
 - Tangible effects (e.g., reduced processing time, improved data accuracy).
 - Intermediate Outcomes
 - Organizational value (e.g., cost savings).
 - Strategic Objectives (Long-term goal)

Benefit Map

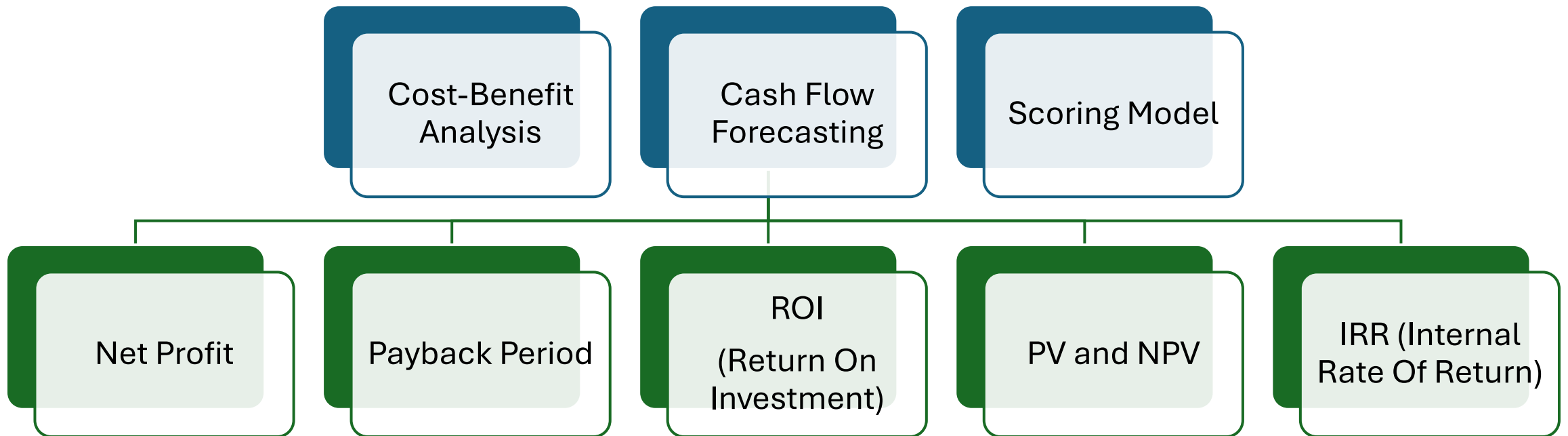


Technical Assessment



- Identifies the functions that the software is expected to perform.
- Evaluate whether the required functionality can be achieved with current or affordable technologies.
- Consider the organizational policy on providing technical infrastructure.
- Align with the SISP (Strategic Information System Plan) of the organization.
 - A Comprehensive document that bridges the technology with business strategy. It defines how information systems (software/IT systems) will help the organization to achieve its goals over the coming years.

Economical Assessment



Cost-benefit Analysis

- Costs
 - Development Cost
 - Fixed/Variable Cost
 - Operation Cost
 - Direct/Indirect Cost
- Benefits
 - Quantified and valued – sales income
 - Quantified but not valued – decrease # no of complaints
 - Identified but not easily quantified – public approval of the organization

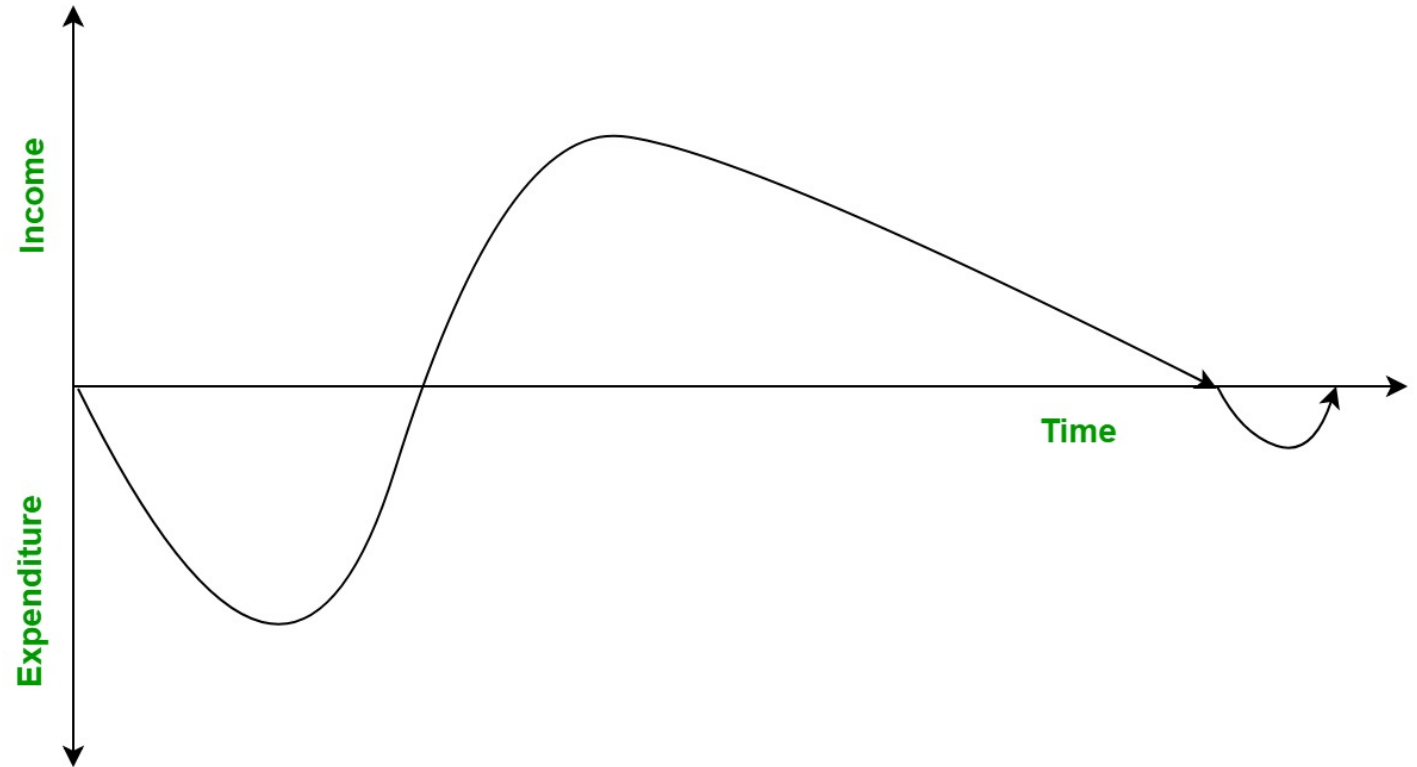
$$\text{NET BENEFIT} = \text{BENEFITS} - \text{COSTS}$$



Cash Flow Forecasting

- What
 - Estimate of the cash inflows and outflows (cash flow) over time
- Why
 - The estimation of the net benefit (previous slide) is not sufficient.
 - Need detailed estimation of benefits and costs versus time.
 - Decision support.

Typical Product Life Cycle Cash Flow



Explanation of Life Cycle Cash Flow

- **Initial Investment Phase** (first downward curve)
 - Shows negative cash flow as money is spent on development, equipment, marketing, and launch costs. This is when you're putting money into the project before seeing returns.
- **Growth Phase** (upward curve)
 - Cash flow becomes positive as the product gains market acceptance and sales increase. Revenue starts exceeding ongoing costs.
- **Maturity Phase** (peak of the curve)
 - Represents maximum profitability when the product has achieved optimal market penetration and efficiency.
- **Decline Phase** (downward slope)
 - Shows decreasing cash flow as sales decline due to market saturation, competition, or product obsolescence.
- **End-of-Life Phase**
 - Represent costs associated with product discontinuation rather than value recovery

Methods of comparing projects using cash flow forecasts

Net Profit

Payback Period

ROI (Return On Investment)

NPV (Net Present Value)

IRR (Internal Rate Of Return)

Net Profit Estimation

Net Profit = Total Revenue – Total Cost

Year	Project 1	Project 2
0	-100,000	-1,000,000
1	10,000	200,000
2	10,000	200,000
3	10,000	200,000
4	20,000	200,000
5	100,000	300,000
Net Profit	50,000	100,000

Payback Period

Year	Net cash flow £	Cumulative balance £
0	(100,000)	(100,000)
1	25,000	(75,000)
2	30,000	(45,000)
3	40,000	(5,000)
4	20,000	15,000

- Calculate the time taken to break even or pay back the initial investment.
- The project with the shortest payback period will be selected.

Payback Period Cont.

Year	Project A	Project B
0	-\$300,000	-\$2,000,000
1	\$100,000	\$600,000
2	\$100,000	\$600,000
3	\$100,000	\$600,000
4	\$100,000	\$600,000
5	\$100,000	\$600,000
Total (Net Profit)	\$200,000	\$1,000,000
Payback Period	3.00 Yrs	3.33 Yrs

$$\frac{200}{600} = \frac{1}{3}$$

ROI (Return On Investment)

- Return on Investment (ROI) is a financial metric used to evaluate the efficiency or profitability of an investment.
- It tells you how much return you're getting compared to the cost of the investment.
- Expressed as a percentage (%).
- Highest ROI will be selected.
- $ROI = (\text{Net Profit} / \text{Investment Cost}) \times 100$.
- Measuring ROI is critical before starting any software project.

Return On Investment (ROI)

$$\text{ROI} = \frac{\text{net profit}}{\text{invested capital (total cost)}} \times 100\%$$



ROI (Return On Investment)

- Example

Development Cost: \$30,000

Expected annual savings in employee time: \$20,000

Tool lifespan: 3 years

Total Benefit = $3 \times 20,000 = 60,000$

ROI = $(60,000 - 30,000) / 30,000 \times 100 = 100\%$

So, you'd get your money back and double the investment over 3 years.

Exercise 01

Net Profit = Total Revenue – Total Cost

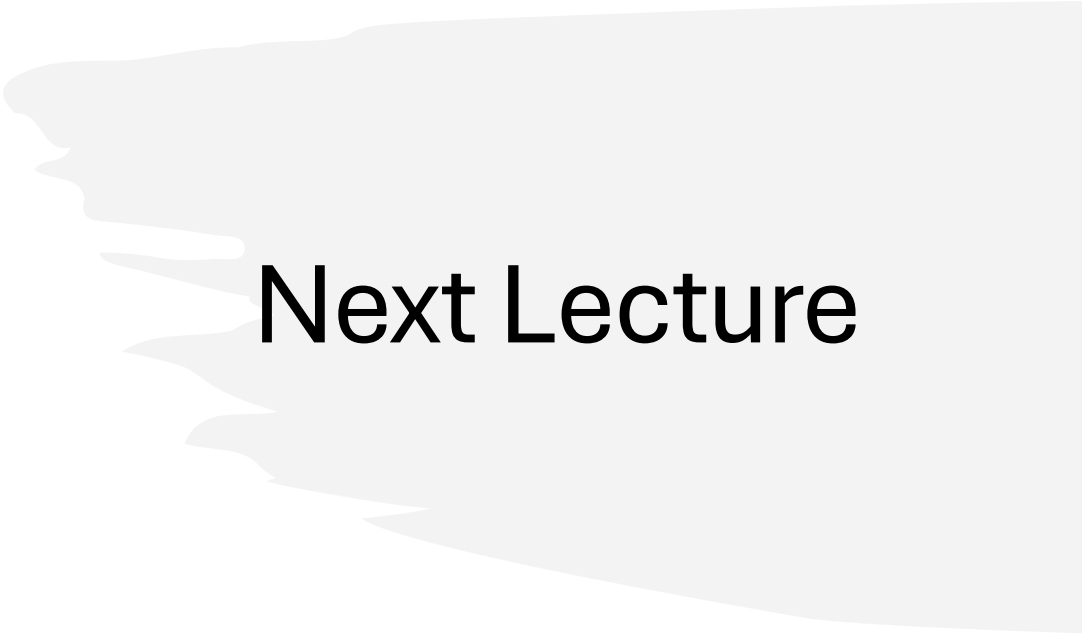
Year	Project 1	Project 2
0	-5,000,000	-4,000,000
1	1,000,000	600,000
2	1,000,000	500,000
3	1,000,000	500,000
4	2,000,000	1,200,000
5	1,000,000	3,000,000
Net Profit	???	???

Exercise 01- Answer

Net Profit = Total Revenue – Total Cost

Year	Project 1	Project 2
0	-5,000,000	-4,000,000
1	1,000,000	600,000
2	1,000,000	500,000
3	1,000,000	500,000
4	2,000,000	1,200,000
5	1,000,000	3,000,000
Net Profit	1M	1.8M

- Present Value and Net Present Value (PV & NPV)
 - Exercises
- IRR (Internal Rate Of Return)
 - Exercises
- Topic 3
 - Project Planning and Scheduling
 - Identify the steps in project planning
 - Identify activities in a given project
 - Different types of activity scheduling techniques
 - Draw activity diagrams



Next Lecture